Multimodal Deepfake Detection with Cross-Modal Attention and Explainable AI

Thilak Raaj N V
Computer Science Department
SRM Institute of science and Technology, Ramapuram
Chennai, India
nt9939@srmist.edu.in

Samuel Raj Irwin V
Computer Science Department
SRM Institute of science and Technology, Ramapuram
Chennai, India
si3076@srmist.edu.in

Balasudhan C M
Computer Science Department
SRM Institute of science and Technology, Ramapuram
Chennai, India
bc0099@srmist.edu.in

Sujatha K
Assistant Professor
Department of CSE
SRM Institute of science and Technology, Ramapuram
Chennai, India
sujathak@srmist.edu.in

Abstract: Advancements in generative AI tools has increased the production and perception of deepfakes, threatening digital trust and information integrity. Dominant methods for detecting deepfakes have used individual modalities that address the problem from a simple binary classification point of view. This has often led to a lack of generalization among different data sets, which cannot capture the complex interactions between audio-visual signals that characterize modern deepfakes. Moreover, these methods are often not transparent since they are based on a black-box principle. In order to address these issues, we have proposed a multimodal deepfake detection framework that classifies deepfake media into different classes based on their manipulation type between the audio and visual signals. Our method involves extracting audio and video frames from the marked altered intervals of the media signals. This has reduced the effects of signal dilution and increased the focus on the altered parts. In order to increase the transparency of the proposed method, we have integrated a Large Language Model that provides clear forensic analysis describing the anomalies that characterize each classification decision. This has provided a generalizable paradigm for digital multimedia forensics.

Keywords: *Deepfake Detection, Multimodal Fusion, Cross-Modal Attention, Explainable AI, Forensics, Audio-Visual Analysis*

I. INTRODUCTION

The term deepfakes, originally created by a Reddit user named Deepfake in 2017 refers to synthetic and highly realistic media content created by artificial intelligence that convincingly impersonates real individuals [2], [8]. Process of producing deepfakes mainly depend on sophisticated deep neural networks such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models [1], [8], [13]. The progressive sophistication in the development and use of AI technology and the proliferation of user-friendly deepfake creation tools have resulted in an exponential rise in the generation of deepfakes [2], [9]. This poses a threat to social, political, personal, and cyber spaces and allows for criminal activities such as identity theft, election manipulation, sophisticated phishing and financial fraud, and the production of pornography. Use of deepfakes has raised ethical and privacy issues that violate the rights of individuals by breaching consent, portraiture rights, and emotional well-being and by eroding public trust in digital media environments [2], [8]. Current generation of deep learning technology is in a continuous process of refinement to minimize visual and auditory errors in the created content. The deepfakes created by such technology have become almost indistinguishable from real content and pose a threat in detecting such content by naked-eye observation and conventional detection techniques [13], [19].

In response to the rising threat, deepfake detection research has exploded with increased number of studies in recent years [2], [8]. Nevertheless, there is still a huge gap in deepfake detection. Most methods focus on a single type and tend to overfit the specific generation method. As a result, these methods fail to generalize and detect unseen and sophisticated deepfakes. Furthermore, deepfake detection methods trained on one dataset perform poorly on other datasets. This demonstrates that the cross-dataset generalization problem remains a pending issue in deepfake detection. With the development of generation technology towards multimodal deepfakes, such as cloned audio and video with lips that do not match, the need for adaptive detection methods increases. The detection methods need to handle complex data representations and perform cross-modal temporal analysis to detect subtle timing errors in the video and audio streams [20].

Also, majority of deepfake detectors that use deep learning techniques tend to be "black boxes," meaning that while a decision is given, no explanation for the decision is provided. This lack of transparency poses number of challenges for forensic analysts and policymakers who require verifiable evidence in determining whether a given media has been doctored. Explainable AI can help make such a system more user-friendly and reliable by being able to show exactly what factors were taken into consideration in arriving at a decision.

II. RELATED WORK

Raza et al. proposed a hybrid model combining transfer learning with custom CNN layers for facial deepfake classification, achieving 94% accuracy and 95% precision on facial datasets, demonstrating that combining pretrained

features with domain-specific layers reduces overfitting [1]. Mubarak et al. conducted a wide-ranging survey of 255 papers across visual, audio, and text deepfake detection, concluding that while high accuracy is achievable on bounded datasets, cross-dataset generalization remains the fundamental unresolved challenge and LLM-generated deepfakes represent an underestimated emerging threat [2].

Abdulhamied et al. leveraged audio-video multimodal fusion using CNN-LSTM audio features and ResNet-50 visual features, achieving AUC 0.97 on matched manipulations but only 0.72 on novel cross-manipulation scenarios, demonstrating both the power and limits of unimodal fusion [3]. Mathews et al. proposed the DFIM-HQ dataset alongside a MesoInception-Net architecture incorporating Grad-CAM for explainability, achieving 99.87% accuracy and establishing XAI as a fundamental requirement for trustworthy forensic models [4].

Sunkari and Srinagesh proposed DeepExplain, a hybrid CNN-LSTM model that identifies frame-level inconsistencies through spatial and temporal analysis with Grad-CAM visualization, achieving 97% accuracy and advocating strongly for transparency in cybersecurity and forensics applications [6]. Bekheet et al. proposed the MAVDD multimodal framework combining facial region analysis with Mel-spectrograms and MFCCs, outperforming all individual modality baselines and confirming that fusion of Xception features with MFCCs yields strong media forensics representations [10].

Zhang et al. proposed AVA-CL, combining cross-modal attention with contrastive learning using InfoNCE loss to reconcile temporal features across face and audio modalities, achieving strong AUC results on FakeAVCeleb and DeepfakeTIMIT while showing that contrastive learning substantially improves cross-modal generalization [15]. Liu et al. developed LipFD for lip-sync deepfake detection using a Global-Region Encoder focusing on semantic-level audio-visual mismatches, achieving 95% accuracy with robustness to common perturbations [16]. Boháček and Farid demonstrated that semantic comparison between audio transcripts and lip-reading outputs can reliably separate real and deepfake videos based on phoneme-facial movement discrepancies [17]. Afchar et al. proposed MesoNet, a compact CNN targeting mesoscopic image features, achieving 98.4% accuracy against Face2Face forgeries and demonstrating that lightweight architectures can capture manipulation artifacts without heavy computation [18]. Raza and Malik proposed MultimodalTrace, combining audio and visual patches through dedicated mixer layers for audiovisual representation learning, achieving state-of-the-art results on FakeAVCeleb [19].

Despite these advances, the literature reveals two persistent gaps:

i. Models trained on a single dataset consistently underperform in cross-dataset evaluation.

ii. Mapping detection decisions to human-interpretable signal-level evidence remains underexplored in multimodal settings. This work directly addresses both.

III. METHODOLOGY

A. Research Design

Rather than training and evaluating on a single dataset, we developed a cross-dataset training and evaluation framework using three publicly available datasets with distinct manipulation types, recording conditions, and audio characteristics. The research is motivated by and addresses three principal questions:

i. Can a single multimodal model generalize across datasets with fundamentally different manipulation types and recording conditions?

ii. Does fine-grained four-class manipulation classification improve forensic utility over binary real/fake detection without sacrificing accuracy?

iii. Can XAI techniques combined with LLM-based report generation provide human-readable forensic explanations that faithfully reflect the model's decision-making process?

B. Datasets and Label Scheme

A unified four-class label scheme is applied across all three datasets, encoding which modality has been altered (Table I) to determine whether a manipulation is limited to the audio track (e.g., a voice clone layered onto genuine video) or limited to the video track (e.g., a face swap applied to genuine audio), rather than simply flagging media as fake. Three publicly available datasets are used, summarized in Table II after preprocessing.

Label	Video	Audio	Description
0	Real	Real	Both modalities are genuine, untampered video and audio
1	Fake	Real	Face swapped or manipulated; audio track untouched
2	Real	Fake	Voice cloned or replaced; video content authentic
3	Fake	Fake	Both face and voice synthetically generated or replaced

TABLE I. UNIFIED FOUR-CLASS LABEL SCHEME

Dataset	Train	Test	Classes	Labels	Key Property
FakeAVCeleb	~17,600	~3,800	4	0–3	Full AV manipulation coverage
LAV-DF	800	320	4	0–3	Temporally localised fakes (~9%)
FF++ C23	640	160	2	0–1	Video-only; silent/absent audio

TABLE II. DATASET SUMMARY AFTER PREPROCESSING

FakeAVCeleb [20] is the primary dataset, containing celebrity videos with four manipulation categories aligned to the label scheme. LAV-DF [15] provides temporally localized fakes where, on average, only approximately 9% of each video contains manipulated content. FaceForensics++ C23

[13] contains six video-only manipulation types; videos without audio tracks receive zero-padded audio arrays to prevent the model from exploiting audio absence as a detection cue.

C. Preprocessing

A consistent preprocessing pipeline is applied across all three datasets. Face detection is performed using RetinaFace, with the bounding box of the largest detected face in the first 30 frames fixed and applied to all subsequent frames. A center crop is used when no face is detected. 16 frames are uniformly sampled per video and resized to 224×224 pixels. Audio is extracted at 16 kHz mono using FFmpeg and divided into 16 mel-spectrogram segments of 128 mel bins and 32-time steps each using the librosa library. Preprocessed data is stored in compressed NumPy (.npz) archives containing the frame tensor, audio tensor, and label. FakeAVCeleb data is split 80/20 by speaker identity to prevent leakage; FaceForensics++ is split 80/20 by manipulation type. For LAV-DF, standard uniform sampling is replaced by timestamp-aware sampling, which concentrates all 16 sampled frames within annotated manipulation intervals.

D. Model Architecture

The proposed architecture (Fig. 1) is a multimodal fusion network with five principal components: a visual encoder, a temporal transformer with frame dropout, an audio encoder, a cross-modal transformer, and an audio-visual synchronization feature.

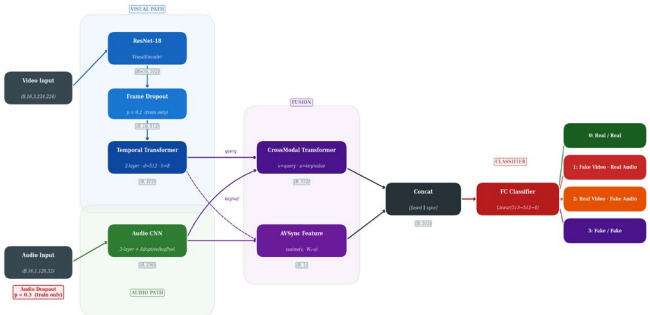

Fig. 1. Complete model architecture. Visual and audio streams are processed independently before fusion through cross-modal attention. The AV Sync scalar provides an explicit desynchronization signal to the final classifier.

Visual Encoder:

A ResNet18 network pretrained on ImageNet with the classification head removed produces a 512-dimensional feature vector per frame. For a batch of B videos with T=16 frames each, all frames are processed simultaneously as a single batch producing output shape (B, T, 512).

Temporal Transformer with Frame Dropout:

During training, the feature sequence is passed through a Temporal Frame Dropout layer that zeros entire frame embeddings with probability p=0.2, preventing over-reliance on any single frame. A 2-layer Transformer Encoder with d_model=512 and 8 attention heads then processes the sequence, capturing long-range temporal patterns such as drifting blending artifacts or boundary flickers. The temporal dimension is averaged to produce a final 512-dimensional video representation.

Audio Encoder:

The audio stream is processed as 16 mel-spectrogram segments, each of shape (1, 128, 32). A 3-layer CNN with max pooling followed by adaptive average pooling extracts 128-dimensional spatial features per segment. These are averaged across segments and linearly projected to a 256-dimensional audio embedding.

Cross-Modal Transformer:

Rather than concatenating visual and audio features directly, a cross-modal attention mechanism is used. The 512-dimensional visual embedding serves as the query; the 256-dimensional audio embedding is projected to 512 dimensions to serve as key and value in an 8-head attention mechanism. This allows visual features to attend selectively to the most relevant audio information, enabling the model to identify temporal misalignments between the two modalities.

AV Sync Feature:

Audio-visual desynchronization is an artifact present in most deepfake generation pipelines, particularly when only one modality is altered. Cosine similarity is computed between the L2-normalized visual embedding and the projected audio embedding, yielding a scalar ranging from -1 (complete desynchronization) to +1 (perfect synchronization). This scalar is concatenated with the fused cross-modal representation before the final linear classifier.

E. Training Configuration

Table III summarizes the training hyperparameters. The model is trained using the Adam optimizer with Focal Loss (γ=2.0) to prioritize difficult misclassified examples and label smoothing (0.1) to reduce overconfidence. Balanced sampling ensures equal class and dataset representation across each training batch. Audio dropout (p=0.3) forces the model to develop a visual-only detection path in addition to the joint multimodal path, improving robustness to missing or corrupted audio.

Parameter	Value	Rationale
Optimizer	Adam (lr=1e-4)	Adaptive learning rate for stable training
Epochs	10	Early stopping on cross-dataset avg accuracy
Loss Function	Focal Loss (γ=2.0)	Focuses learning on hard misclassified examples
Label Smoothing	0.1	Reduces overconfidence; improves calibration
Gradient Clipping	Max norm 1.0	Prevents exploding gradients in transformer layers
Audio Dropout	p = 0.3	Forces model to learn video-only detection path
Frame Dropout	p = 0.2	Temporal regularisation during training
Mixed Precision	AMP (fp16)	2× memory efficiency; faster training

TABLE III. TRAINING HYPERPARAMETERS

F. XAI and Forensic Report Generation

Three complementary XAI techniques generate evidence at different levels of granularity. Grad-CAM produces spatial activation maps over video frames, highlighting image regions that contributed most to the classification decision. Temporal attention weights from the Transformer Encoder identify which frames within the sequence were most influential. Audio saliency maps highlight frequency-time regions in the mel-spectrogram most relevant to the decision. These three evidence streams are structured into a natural language prompt and submitted to an LLM, which generates a human-readable forensic report describing detected anomalies and their significance. This pipeline converts an otherwise opaque neural network decision into verifiable, auditable evidence suitable for forensic and legal contexts.

IV. RESULTS AND DISCUSSION

A. Training Dynamics

Training loss decreased consistently across epochs, with rapid reduction in the initial epochs as the model acquires fundamental audio-visual representations, followed by a more gradual convergence as it fine-tunes decision boundaries (Fig. 2). Training accuracy improves smoothly throughout without significant oscillations, indicating stable optimization free from severe overfitting or divergence (Fig. 3). The absence of abrupt fluctuations in both curves confirms that the multi-dataset balanced sampling strategy and regularization components—frame dropout, audio dropout, and gradient clipping—work together to stabilize training.

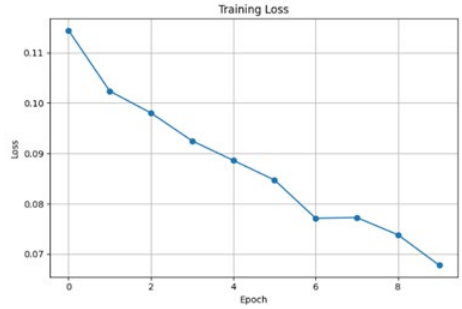

Fig. 2. Training loss curve showing consistent convergence.

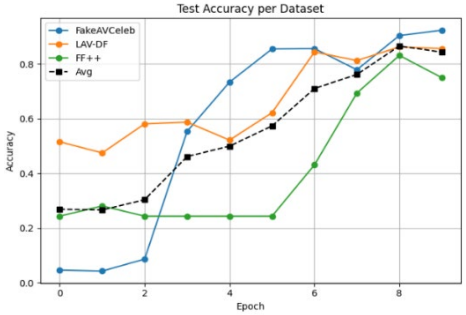

Fig. 3. Training accuracy vs. epoch showing stable improvement.

B. ROC and DET Analysis

ROC curves (Fig. 4) quantify the model's discriminative capability across classification thresholds. The model achieves macro-AUC of 0.980 on FakeAVCeleb, demonstrating near-perfect separation between real and manipulated samples across all four manipulation classes. On LAV-DF, macro-AUC of 0.949 confirms strong detection of temporally localized fakes despite the relatively small fraction of manipulated content per video. On FaceForensics++, macro-AUC of 0.852 reflects the inherent difficulty of detecting manipulations in highly compressed video rather than an architectural limitation.

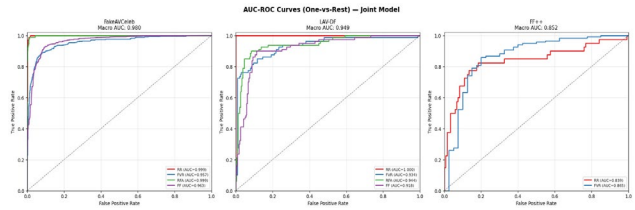

Fig. 4. ROC-AUC curves across all three evaluation datasets.

DET curves (Fig. 5) provide complementary analysis through Equal Error Rates (EER). The model achieves EER of 0.9% on FakeAVCeleb and 0.0% on LAV-DF, indicating virtually perfect separation at optimal thresholds. The EER of 17.9% on FaceForensics++ is consistent with the compression artifacts present in C23-quality video, which attenuate the high-frequency manipulation traces that visual encoders typically exploit.

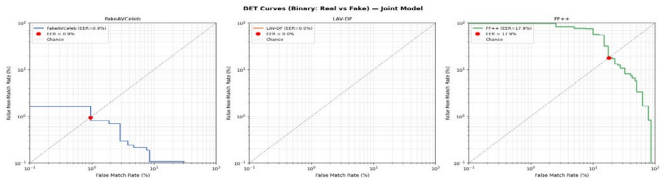

Fig. 5. DET curves (binary real vs. fake) across datasets.

C. Cross-Dataset Generalization

Per-class F1 scores (Fig. 6) confirm balanced detection performance across all four manipulation types. On FakeAVCeleb, the model achieves F1 scores of 92.9% for Real/Real, 89.3% for Fake Video/Real Audio, 89.6% for Real Video/Fake Audio, and 91.2% for Fake/Fake, demonstrating consistently strong performance across all classes without category-specific collapse. On LAV-DF, the model achieves an F1 score of 99.4% for authentic samples and above 82% for all manipulated categories, confirming effective detection of the localized manipulation signal identified by timestamp-aware preprocessing. On FaceForensics++, which presents only two classes, F1 scores of 69.0% for Real/Real and 88.9% for Fake Video/Real Audio highlight that the primary challenge is avoiding false positives on real samples in a high-compression environment.

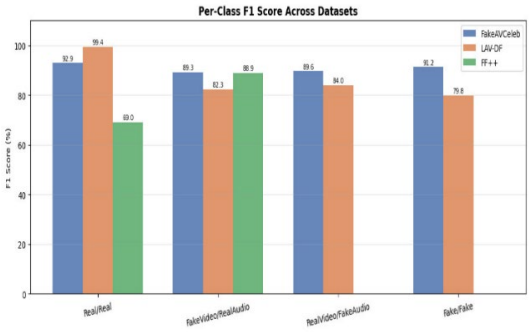

Fig. 6. Per-class F1 scores across all datasets.

The cross-dataset DET comparison (Fig. 7) further illustrates the generalization capability of the proposed architecture. The curves reveal clear differences in dataset difficulty while confirming that the model maintains relatively low error rates across manipulation scenarios it was not specifically tuned for. This generalization arises from the joint cross-dataset training strategy, which exposes the model to diverse manipulation types simultaneously, preventing the statistical overfitting to a single generation method that limits most prior systems.

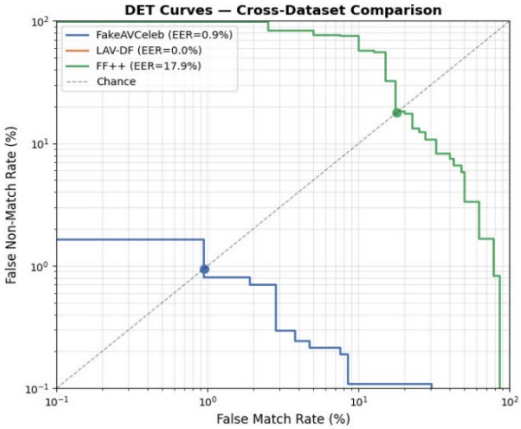

Fig. 7. Cross-dataset DET comparison illustrating generalization across manipulation types.

D. Prediction Confidence

The distribution of maximum softmax probabilities (Fig. 8) reveals that correctly classified samples consistently exhibit higher confidence scores than misclassified ones across all three datasets. On FakeAVCeleb, most correct predictions cluster in the 0.50–0.65 probability range, while incorrect predictions are concentrated at lower values. Similar patterns appear on LAV-DF and FaceForensics++. This separation between correct and incorrect prediction confidence distributions is practically significant: it enables the deployed system to flag low-confidence predictions for human expert review, creating a human-in-the-loop workflow appropriate for high-stakes forensic applications.

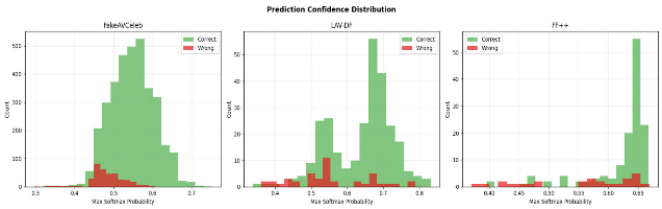

Fig. 8. Prediction confidence distribution, separating correct (higher confidence) from incorrect predictions.

Overall, the proposed framework achieves 90.4% accuracy on FakeAVCeleb, 86.3% on LAV-DF, and 83.1% on FaceForensics++, yielding an average cross-dataset accuracy of 86.6%. These results confirm that joint multimodal training across heterogeneous datasets produces a detector with both strong discriminative performance and meaningful generalization.

V. Future Scope

While overall performance is good, making it easier to generalize to highly compressed and unseen deepfake data is still a big problem. To address this, future research should focus on compression aware training methodologies and integrating more resilient visual encoders, such as CLIP-based architectures, to stabilize feature representation across diverse video qualities. It is essential to fix this problem to ensure that the detection systems work well in the real world, where media often goes through many stages of compression and transformation.

VI. Conclusion

Combining spatial, temporal, and audio-based features, made the system works well with a lot of different datasets, with an average accuracy of 86.6% and high AUC scores. Equal Error Rates are low on most datasets indicating that the system can differentiate between real and fake samples very well. The balanced F1-scores also show that the system can find samples that have been changed in different ways. The combination of multimodal features is better and more flexible than unimodal methods.

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